GREAT POWER

# ESS REFERENCE PROJECTS

World-Class Lithium-Ion Battery Manufacturer

# CONTENTS

01 / KEY PROJECTS

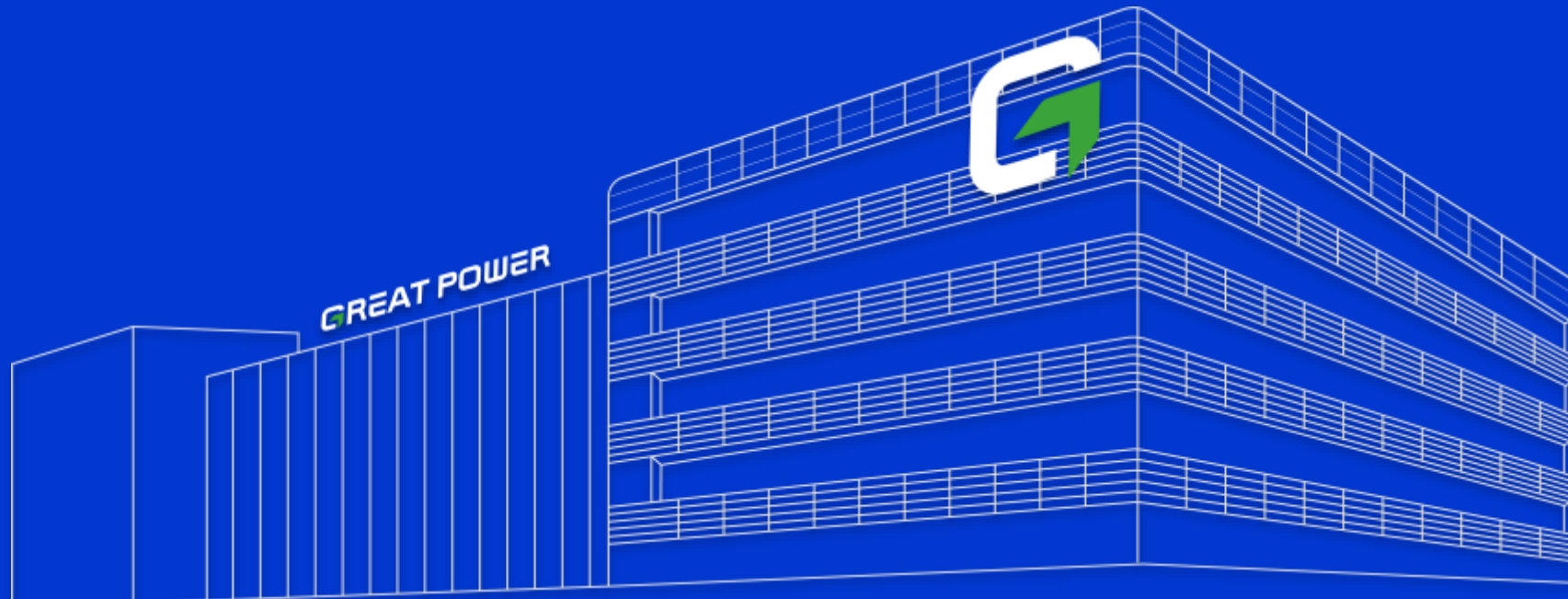
02 / INTERNATIONAL PROJECTS

03 / CHINA LARGE-SCALE PROJECTS

04 / CHINA C&I PROJECTS

# 01

## KEY PPOJECTS



# International Projects Overview

As a world-class lithium-ion battery manufacturer, Great Power has deployed energy storage projects in more than 50 countries and regions, maintaining a zero-incident safety record across its portfolio. In 2025, its ESS shipments surpassed **2 GWh**, earning **BloombergNEF Tier 1 recognition** for the seventh consecutive time. With deep technical expertise, tailored solutions, and fast, reliable execution, Great Power continues to earn the confidence of customers worldwide.

**India ESS Project**
Scale: 722MWh
Total 4GWh, 722MWh already shipped.

**Azerbaijan National-Level ESS Project**
Scale: 500MWh

**Lithuania ESS Project**
Scale: 180MWh

**U.S. Virginia ESS Project**
Scale: 150MWh

**Belgium ESS Project**
Scale: 17.5MWh

**Japan, Yokohama School PV + ESS Project**
Scale: 17.5MWh

**Germany Containerized ESS Project**
Scale: 15MW/16.77MWh

**Netherlands PV + ESS Hybrid Project**
Scale: PV 12.5MWp，ESS 3.6MW/3.6MWh

**India Off-Grid PV + ESS Project**
Scale: 1MW/2MWh

**Croatia ESS Project**
Scale: 500kW/1MWh

# China Large Scale Projects Overview

As a world-class lithium-ion battery manufacturer, Great Power has deployed energy storage projects in more than 50 countries and regions, maintaining a zero-incident safety record across its portfolio. In 2025, its ESS shipments surpassed **2 GWh**, earning **BloombergNEF Tier 1 recognition** for the seventh consecutive time. With deep technical expertise, tailored solutions, and fast, reliable execution, Great Power continues to earn the confidence of customers worldwide.

**Sichuan Zhongfu ESS Project**
Scale: 100MW/400MWh

**Yunnan Xiangyun Shared ESS Project**
Scale: 200MW/400MWh

**Jiashi PV Project**
Scale: 150MWh/600MWh

**Gansu Wuwei Standalone ESS Project**
Scale: 105MW/420MWh

**Tumushuke PV-ESS Project**
Scale: 87.5MW/350MWh

**Bachu Agrivoltaic + ESS Project**
Scale: 35MW/140MWh

**Jimsar PV + ESS Project**
Scale: 37.5MW/150MWh

**Guangxi Nanning ESS Project**
Scale: 50MW/100MWh

**Jiangsu Shipyard ESS Project**
Scale: 30MW/100MWh

**Inner Mongolia Ulanqab Green ESS Project**
Scale: 90MW/180MWh

# China C&I Projects Overview

As a world-class lithium-ion battery manufacturer, Great Power has deployed energy storage projects in more than 50 countries and regions, maintaining a zero-incident safety record across its portfolio. In 2025, its battery shipments surpassed **20 GWh**, earning **BloombergNEF Tier 1 recognition** for the seventh consecutive time. With deep technical expertise, tailored solutions, and fast, reliable execution, Great Power continues to earn the confidence of customers worldwide.

**Shenzhen Bus Public Charging Station ESS Project**
Scale: 30MW/60MWh

**Zhejiang Xianhe New Energy ESS Project**
Scale: 44.72MWh

**Zhongyuan Neipei Group Smart Energy ESS Project**
Scale: 40MWh

**Henan Jinma Energy Smart ESS Project**
Scale: 34.4MWh

**Anyang Shenlong Smart ESS Project**
Scale: 34.4MWh

**Henan Mingsheng New Materials ESS Project**
Scale: 30.96MWh

**Zhongshan Zhongjing Semiconductor Smart ESS Project**
Scale: 27.52MWh

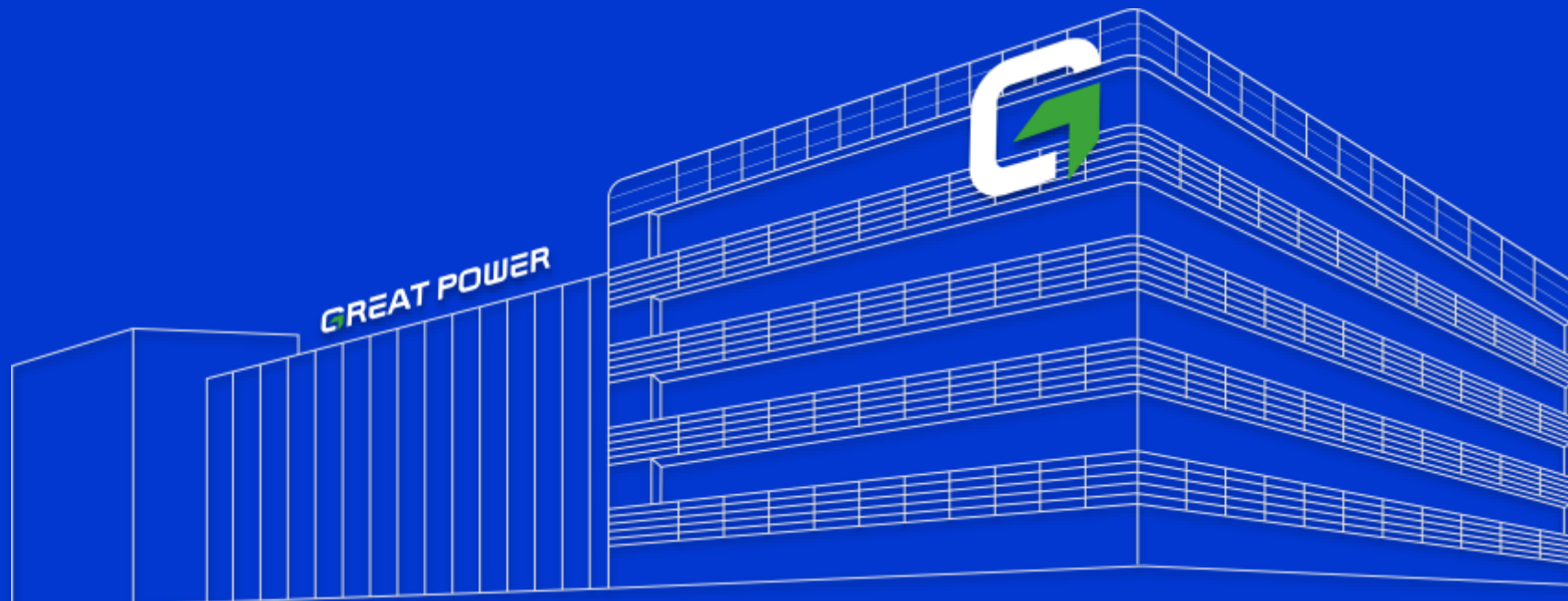
**Hunan Anfu Environmental ESS Project**
Scale: 25MWh

**Henan Jinli Jinqian Smart ESS Project**
Scale: 24.08MWh

**Great Power Zhuhai Industrial Park ESS Project**
Scale: 29.77MWh

# 02

# INTERNATIONAL PROJECTS



GREAT POWER

# India - 722MWh Energy Storage Project

**Sector: Energy & Utilities**

**Scale:** Total 4GWh, 722MWh already shipped.

## Project Overview

Great Power is participating in the Khavda Renewable Energy Park, the world’s largest single-location renewable energy project. Phase I totals 4 GWh, with 722 MWh already shipped, while Phase II (8,840 MWh) is currently under agreement. The project is expected to scale up to 30 GWh by 2029. As a key supplier, Great Power provides 5 MWh DC-side containerized energy storage systems (overseas version), supporting large-scale renewable integration with high efficiency and reliability.

## Project Value

This project marks a major milestone in Great Power’s participation in world-leading giga-scale renewable energy developments, reinforcing its capability in ultra-large utility projects. It demonstrates the scalability, reliability, and adaptability of Great Power’s energy storage solutions in complex hybrid energy systems, supporting efficient integration of solar and wind power. The project further strengthens Great Power’s position as a global supplier for large-scale energy storage, contributing to long-term energy transition aligned with the project’s 30 GWh vision.

# Azerbaijan - 500MWh National-Level ESS Project

**Sector: Energy & Utilities**

**Scale:** 500MWh

## Project Overview

The 500 MWh YEO energy storage project in Azerbaijan is the largest in the country, owned by Azerenergy OJSC. Delivered under a turnkey EPC model, it covers battery systems, PCS, BMS, EMS, and full grid integration. Once completed, the project will connect to multiple substations and provide key services such as frequency regulation and black start. Great Power supplies 5 MWh DC-side containerized systems, supporting a more stable and flexible power system.

## Project Value

Serving as the largest energy storage project in Azerbaijan, it highlights the growing role of utility-scale storage in national energy infrastructure and grid modernization. The project demonstrates strong capability in delivering integrated, large-scale energy storage solutions for government-backed utilities, with high compatibility across complex grid environments. By enabling advanced applications such as frequency control and black start, it enhances grid reliability and supports renewable energy integration. It also reinforces Great Power’s competitiveness in international markets, showcasing its ability to deliver scalable solutions across diverse regions.

# Lithuania - 180MWh Energy Storage Project

**Sector: Energy & Utilities**

**Scale:** 180MWh

## Project Overview

Jointly contracted by Great Power and Connecto, this project is currently the largest energy storage installation in the Baltic region. The project is under continuous delivery, with grid connection scheduled for Q2 2026. As Great Power’s largest project in Europe to date, the system integrates European PCS and EMS platforms and is delivered under DDP terms, enabling participation in power market trading and a range of ancillary grid services.

## Project Value

This project represents a significant milestone in Great Power’s expansion into Europe’s utility-scale energy storage market, strengthening its presence in grid-side applications across the region. It demonstrates the high compatibility and multi-functional adaptability of Great Power’s solutions for electricity market participation and grid support, while also showcasing the company’s ability to provide localized European services and comprehensive project delivery.

# Virginia, USA - 150MWh Data Center Backup

**Sector: Commercial & Logistics**

**Scale:** 150MWh（30 Ultra Max 5000 Energy Storage Containers）

## Project Overview

The client is a leading distributor with extensive warehousing, logistics, and cold chain infrastructure, which relies on uncompromised data center power stability to maintain business continuity. To address this need, Great Power delivered a tailored integrated AC/DC solution optimized for load shifting. The 50MWh Phase 1 deployment in Virginia is now operational, enabling rapid commissioning and seamless grid integration.

## Project Value

By integrating Great Power’s advanced energy storage system, the client now benefits from a reliable, rapidly deployable backup power source that effectively mitigates grid outages and voltage fluctuations. This ensures business continuity and enables the company to consistently deliver on its commitments to customers.

# Puerto Rico, USA - 2.5MWh Pharma Backup

**Sector: Healthcare**

**Scale:** 1.25MW/2.5MWh (6 Magna-Series 418kWh Outdoor Energy Storage Cabinets)

## Project Overview

The client is a global leader in pharmaceuticals—requires ultra-safe, efficient, and long-lasting backup power for its Puerto Rico production facility. Great Power supplied 6 Magna-Series 418kWh outdoor cabinets, which deliver 12% higher energy density than standard industry single-rack units. These energy storage cabinets will play a crucial role in integrating renewable energy and providing backup power for critical resources on the island, and offers a 20-year service life.

## Project Value

The 418kWh cabinets are powered by Great Power’ s in-house 314Ultra cells, with fully Designed, integrated, and manufactured DC modules (end-to-end control of core components). The system is fully certified to UL 9540 and compliant with NFPA 855, ensuring the highest levels of safety and regulatory compliance for mission-critical applications in the healthcare sector.

# San Diego, USA - 373KWh R&D Project

**Sector: Research & Education**

**Scale:** 125kW/373kWh (1× 373kWh Outdoor Energy Storage Cabinets)

## Project Overview

Great Power delivered a 125 kW / 373 kWh energy storage demonstration system for a world-class research university, specifically designed for academic research applications. The system will support studies in campus renewable integration and power management, providing reliable and stable electricity for laboratories and research facilities. This pilot project serves as a proving ground for evaluating the real-world performance of energy storage technologies within academic settings and campus microgrids.

## Project Value

We are pleased to support a world-class research university with our advanced energy storage solution. The system utilizes Great Power’s self-developed, high-safety LFP batteries, offering exceptional cycle life and energy efficiency while meeting all relevant U.S. safety standards. This project represents an important demonstration for Great Power in the research and education sector and provides critical infrastructure for the campus’s sustainability initiatives, supporting innovation, carbon-reduction goals, and long-term energy resilience across campus.

Location: Croatia, Sisak

Scale: PV 2.76MWp, ESS 16.56MWh

**Project Overview**

Certification: TUV PP59044A

**Project Value**

Tailored PV + ESS solution for industrial applications, enabling hybrid PV-storage integration, peak-shaving, load balancing, and carbon trading, while delivering safe, efficient, and reliable energy.

Location: Switzerland

Scale: PV 12.5 MW, ESS 3.6 MW / 3.6 MWh

**Project Value**

Switzerland has summer electricity surplus but imports power in winter. The client required high performance and long lifespan. After multiple design iterations, the system was successfully grid-connected and recognized for its efficiency and economic benefits.Great Power provides containerized storage to optimize seasonal power distribution, enabling peak shaving, frequency regulation, dynamic expansion, and backup power.

Location: Netherlands, Hattemerbroek

Scale: ESS 5MW/6MWh

**Project Value**

Energy storage for EV charging stations. A milestone project for electric trucks and charging infrastructure in the Netherlands. Delivered in collaboration with Swiss BESS partners and Dutch PV + ESS developers. BESS provides 90% of the energy to charge EV trucks overnight using stored PV energy.

Location: Croatia

Scale: 500kW/1MWh

**Project Overview**

COD：2017 (stable operation for 9 years)

Certification: TUV approved

**Project Value**

Great Power’s first user-side project in Europe, addressing peak electricity demand at hotels and grid instability, enabling peak-valley arbitrage and backup power, establishing a strong European market reputation.

Location: India

Scale: 1MW/2MWh

**Project Overview**

Off-grid PV + ESS + Diesel Hybrid Microgrid Project

**Project Value**

Located in India’ s tropical monsoon region, this off-grid microgrid combines solar PV, modular energy storage, and diesel generation to provide reliable backup power for a village. The area experiences heavy seasonal rainfall and extreme temperatures up to 40°C, and the system is designed to ensure continuous, efficient, and safe operation under these challenging conditions.

Location: Japan, Yokohama

Scale: PV 6.5MWp，15.08MWh

**Project Value**

Project Value: Distributed across 65 schools, the system maximizes PV utilization, reduces electricity costs, and maintains backup off-grid capability when PV surplus is insufficient.

Location: Germany, Wismar

Scale: PV 15MWp, ESS 16.77MWh

**Project Overview**

Typical PV + ESS application in Europe. Supports load smoothing, peak-valley arbitrage, frequency regulation, and backup power, delivering both economic and social bene

Location: Eemnes, Netherlands

Scale: PV 12.5 MWp, ESS 3.6 MW / 3.6 MWh

**Project Value**

Tailored ESS solution for customer-specific industrial needs. Enables peak-shaving, frequency regulation, dynamic expansion, and backup power, enhancing clean energy utilization efficiency.

Location: Belgium

Scale: 17.5MWh

**Project Overview**

Client：OmniPower B.V

COD：2025

Equipment: 56 × 313 kWh outdoor cabinets

Location: Netherlands

Scale: 9.5MWh

**Project Overview**

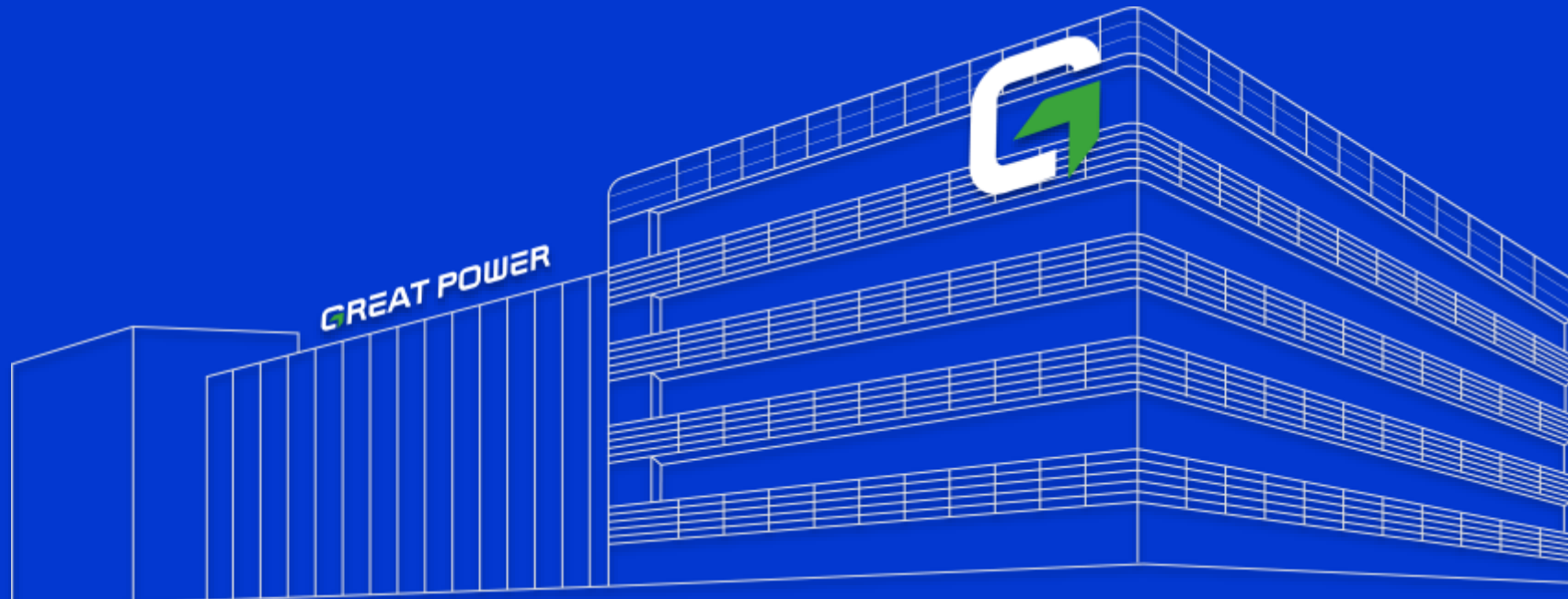
Client：Intercel B.V.

COD：2025

Equipment: 2 × 3.44 MWh containers + 7 × 280 kWh outdoor cabinets + 3 × 233 kWh outdoor cabinets

# 03

## CHINA LARGE-SCALE PROJECTS



GREAT POWER

# Guangyuan - 400MWh Energy Storage Project

**Sector: Energy & Utilities**

**Scale:** 100MW/400MWh

## Project Overview

Jointly developed by Great Power and Sichuan Zhongfu, this project is currently the largest user-side lithium-ion energy storage installation in China. Located in the Yuanjiaba Industrial Park of Guangyuan Economic and Technological Development Zone, the 100 MW/400 MWh system serves the aluminum electrolysis industry, marking the first large-scale user-side storage deployment in this sector. The project features a pioneering DC storage connection technology specifically designed for the aluminum industry and integrates it with solar PV to create an innovative "PV + storage" DC direct-supply system.

## Project Value

The project sets a new benchmark for energy efficiency and cost-effectiveness in the aluminum electrolysis sector by significantly reducing power losses and enabling advanced energy management. It demonstrates Great Power's ability to deliver industrial-scale, industry-specific storage solutions while supporting low-carbon industrial development. The project also contributes to the green transformation of Guangyuan Economic and Technological Development Zone, advancing the creation of a modern, low-carbon power system.

GREAT POWER

# Dali - 400MWh Shared Energy Storage Project

**Sector: Energy & Utilities**

**Scale:** 200MW/400MWh

## Project Overview

Located in Xiangyun County Industrial Park in Dali, Yunnan Province, this 200 MW/400 MWh independent shared energy storage power station is now in stable operation. As one of the largest single-site energy storage projects in Yunnan, the project is equipped with Great Power' s DC-side lithium iron phosphate (LFP) battery storage system, along with an intelligent operations platform designed to enhance system reliability, durability, and overall project returns.

## Project Value

The project efficiently absorbs renewable energy during peak wind and solar generation periods and provides stable power during peak industrial demand, enabling effective peak shaving and load shifting while smoothing fluctuations in renewable generation. Through its shared energy storage model, the project also flexibly coordinates the needs and resources of power generation, the grid, and end users, supporting the development of a more efficient and resilient regional power system.

GREAT POWER

# Jiashi - 600MWh PV Project

**Sector: Energy & Utilities**

**Scale:** 200MW/400MWh

## Project Overview

Located in Jiashi County, this 600 MWh PV-plus-storage project was developed in collaboration between Great Power and Jichai Power Company Limited of China National Petroleum Corporation (CNPC). Great Power supplied 3.35 MWh air-cooled walk-in containerized energy storage systems with a 1500 V DC architecture, providing reliable storage support for the integrated solar generation system.

## Project Value

Designed to support high-energy-demand industrial operations in the oil sector, the project enhances the efficiency and stability of solar power generation through integrated energy storage. The system generates approximately 1.04 TWh of electricity annually, equivalent to replacing about 312,000 tons of standard coal and reducing $CO_2$ emissions by roughly 812,000 tons each year, along with significant reductions in $SO_2$ and NOx emissions. The project helps improve clean energy utilization at the Tarim Oilfield while strengthening power system stability and supporting a cleaner, more efficient power supply in the Kashgar region.

GREAT POWER

# Ulanqab - 420MWh Green Energy Storage Project

**Sector: Energy & Utilities**

**Scale:** 105MW/420MWh

## Project Overview

Located in Chaoyang Town, Huade County, Ulanqab, Inner Mongolia Autonomous Region, this 105 MW/420 MWh independent energy storage project represents a hybrid storage demonstration integrating multiple technologies. The project is innovatively coupled with a compressed air energy storage (CAES) facility, forming a hybrid system that combines large-capacity, long-duration storage with fast-response electrochemical energy storage to support diverse grid applications.

## Project Value

The project facilitates the grid integration and efficient utilization of renewable energy in Ulanqab and surrounding regions, helping to reduce reliance on fossil fuel generation. It can also serve as an emergency backup power source for the regional grid, responding rapidly during grid faults or peak demand periods to provide power support and enhance the reliability and resilience of electricity supply in Huade County and nearby areas.

Tumxuk Guangdong Electric Hanhai PV-Plus-Storage Project (2024)

Scale: 87.5MW/350MWh

**Project Overview**
Located in Tumxuk, the project integrates PV generation with a 350 MWh energy storage system using coordinated PV-storage control. It also incorporates ecological restoration beneath the PV arrays together with windbreak and sand stabilization measures.

**Project Value**
The project supplies clean electricity, reduces carbon emissions, enhances grid stability, and supports local employment and desert ecosystem restoration.

Bachu Agrivoltaic-Plus-Storage Project (2024)

Scale: 35MW/140MWh

**Project Overview**
Located in Bachu, this agrivoltaic project integrates solar power with livestock farming. Great Power supplied high-quality battery cells to smooth wind generation fluctuations and enhance local power infrastructure.

**Project Value**
The project promotes clean energy adoption, aligns renewable power and agricultural interests, and supports a sustainable modern pastoral economy.

Jimusaer PV-Plus-Storage Project (2024)

Scale: 37.5MW/150MWh

**Project Overview**
Located in Jimusaer, this PV-plus-storage project covers 5,558 acres with 48 subarrays and a 37.5 MW / 4-hour storage system. Great Power supplied the core battery cells.

**Project Value**
The project smooths solar generation, reduces curtailment, and strengthens renewable energy supply and grid support.

Jiangsu Shipyard Energy Storage Project

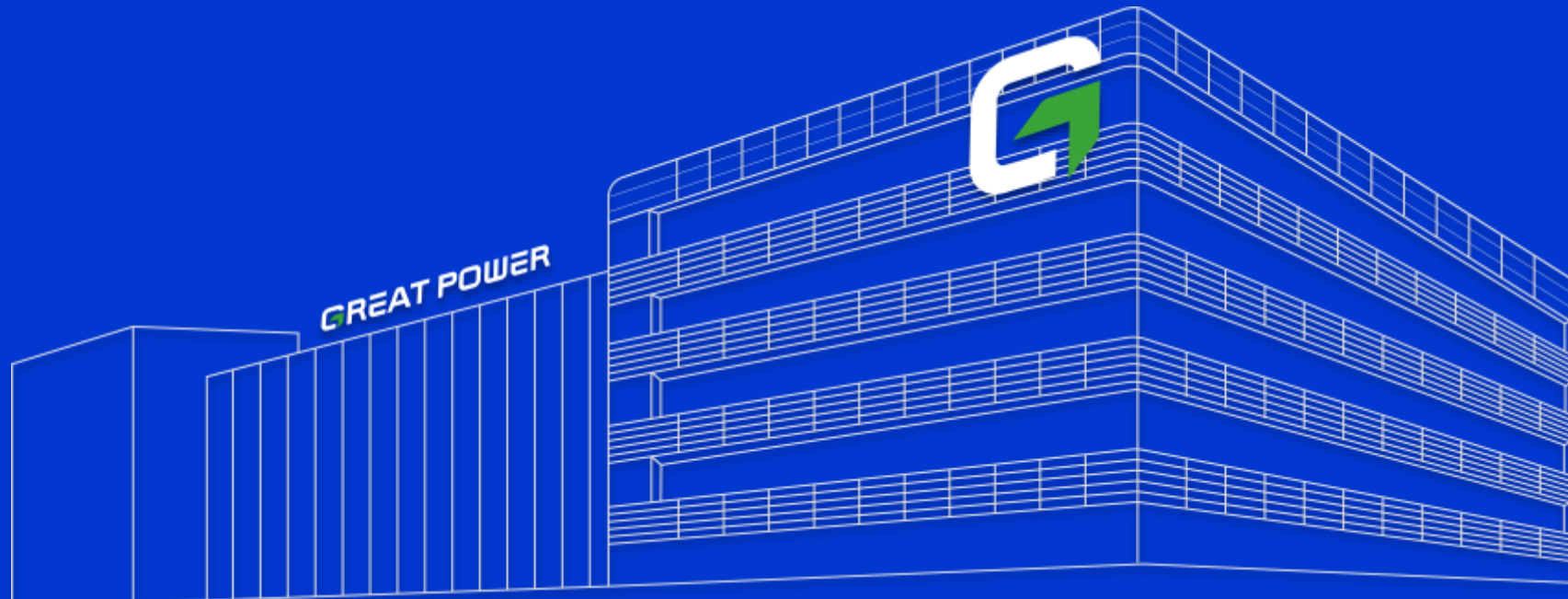
Scale: 30MW/100MWh

**Project Overview**
Located at a state-owned shipyard in Jiangsu, this industrial energy storage project supports high-power, continuous operations. Great Power supplied five energy storage subsystems to meet large equipment startup and load demands.

**Project Value**
The project provides stable clean power, enables peak shaving and load shifting, supports dynamic load management, and helps the shipyard advance low-carbon operations while enhancing Jiangsu’s green electricity capabilities.

# 04

## CHINA C&I PROJECTS



GREAT POWER

# Shenzhen - 60MWh Bus Public Charging Station ESS Project

**Sector: Energy & Utilities**

**Scale:** 30MW/60MWh

## Project Overview

Jointly developed by Great Power, Shenzhen Bus Group, and Intellifusion, this project is the largest distributed commercial and industrial (C&I) energy storage deployment in China as of 2023. The project deploys 280 units of Great Power' s Great One 200 systems, totaling 30 MW / 60 MWh across 33 bus depots in Shenzhen, supporting the development of a "New Energy Intelligent Bus

## Project Value

Spanning 33 sites across Shenzhen, the project demonstrates the scalability and reliability of Great Power' s C&I energy storage solutions in real-world urban applications. As a flagship deployment of the "New Energy Intelligent Bus Ecosystem," it highlights strong customer recognition of Great Power' s products and services while supporting cleaner, smarter energy use in urban transportation systems.

GREAT POWER

# Quzhou - 44.72MWh Energy Storage Project

**Sector: Energy-Intensive - Paper Industry**

**Scale:** 44.72MWh

## Project Overview

Located in Quzhou, Zhejiang Province, this project serves Xianhe Co., Ltd., a major listed paper manufacturer in China with an integrated industrial chain and annual capacity exceeding 500,000 tons. As a typical user-side commercial and industrial energy storage deployment in the paper industry, the project demonstrates Great Power’ s storage solutions in high-energy-demand manufacturing scenarios.

## Project Value

Papermaking processes such as pulping, mixing, and drying require significant and stable power supply. Great Power provides customized energy storage solutions for Xianhe’ s industrial parks nationwide, enabling peak-valley price arbitrage, providing backup power, and supporting peak shaving and load shifting based on grid demand. The project helps ensure reliable production power while reducing operational costs.

GREAT POWER

# Mengzhou - 40MWh Smart Energy Storage Project

**Sector: Energy & Utilities**

**Scale:** 40MWh

## Project Overview

Deployed by Great Power for Zhongyuan Internal Fittings Group in Henan Province, this project delivers a total energy storage capacity of 40.105 MWh, comprising 6 GreatCom-20HC-5000 liquid-cooled energy storage containers and 47 Great One-200 storage cabinets. Designed for industrial applications, the system provides reliable on-site energy storage to support stable power supply and efficient energy management for large-scale manufacturing operations.

## Project Value

The project helps reduce energy costs through peak shaving and load shifting while increasing the share of green electricity to support the company' s low-carbon transition. It enhances power supply reliability, safeguards continuous production, and improves energy efficiency through digitalized management, ultimately strengthening the enterprise' s economic, environmental, and social performance as well as its market competitiveness.

GREAT POWER

# Zhenzhou - 30.96MWh New Materials Energy Storage Project

**Sector: Energy-Intensive - Alloy & Advanced Materials**

**Scale:** 30.96MWh

## Project Overview

Deployed by Great Power for Henan Mingsheng New Materials Technology, this project represents the largest commercial and industrial energy storage deployment in the new materials sector in Central China. The system installs 9 GreatCom-20HC-3440 energy storage containers with a total capacity of 30.96 MWh. Designed for precision alloy manufacturing, the project integrates an intelligent energy storage system with high-precision production processes such as rolling and heat treatment, delivering stable, around-the-clock power through millisecond-level load response and coordinated multi-energy control.

## Project Value

As the first energy storage integration solution in China tailored for precision alloy manufacturing, the project overcomes ultra-high-power instantaneous charge-discharge challenges with a system response time of ≤50 ms. Leveraging an "AI prediction + dynamic optimization" strategy, the project saves more than RMB 8 million in electricity costs annually while reducing overall energy consumption per unit of product by 22%, supporting both cost efficiency and low-carbon industrial development.

Henan Jinma Energy Smart Energy Storage Project

Scale: 34.4MWh

**Project Overview**
Deployed for Jinma Energy, a Hong Kong-listed steel–coke enterprise in China, this project is one of the leading C&I energy storage deployments in Henan. It uses Great Power’ s GreatCom liquid-cooled storage solution to support high-energy-demand industrial operations.

**Project Value**
The project reduces electricity costs through peak shaving, enhances power supply reliability, and supports stable grid operation, contributing to regional low-carbon development.

Anyang Shenlong Smart Energy Storage Project

Scale: 10 × GreatCom-20HC-3440, 34.4 MWh

**Project Overview**
Deployed by Great Power for Anyang Shenlong Tengda New Materials in Henan, this project supports peak load management and emergency power supply for high-energy-demand metal new materials production.

**Project Value**
The project ensures stable production, reduces outage risks, and provides a replicable solution for low-carbon transformation in energy-intensive industries.

Zhongshan Zhongjing Semiconductor Smart Energy Storage Project

Scale: 8 × GreatCom-20HC-3440, 27.52 MWh

**Project Overview**
Deployed by Great Power for Zhongjing Semiconductor in Zhongshan, this C&I energy storage project installs 8 GreatCom-20HC-3440 storage containers with a total capacity of 27.52 MWh to support efficient energy management for semiconductor manufacturing.

**Project Value**
The project reduces electricity costs through peak–valley price arbitrage while providing long-term energy savings, serving as a strong demonstration of large-scale user-side energy storage applications.

Hunan Anfu Environmental Energy Storage Project

Scale: 5 × GreatCom-20HC-5000, 25 MWh

**Project Overview**
The PV-plus-storage system prioritizes daytime industrial power use and shifts excess energy to the grid, enabling peak–valley price arbitrage.

**Project Value**
The project boosts economic and environmental benefits, improves energy reliability, and showcases a low-carbon, efficient industrial energy solution.